

Table 2: Popular Web parsers

Web parser	Programming platform
htmlPurifier	PHP
WebGear	Perl
HtmlCleaner	Java
The PHP DOMDocument-class	PHP
Tooska	C++
Lexbor (HTML module)	C
html5lib	Python and PHP
HTML::Parser	Perl
jsdom	JavaScript
Jaunt API	Java
Aspose.HTML	C#
BrilliantHTML5Parser	Swift 3
JTidy	Java
NekoHTML	Java
Lambda Soup	OCaml
Hubbub	C
libxml2 HTMLparser	C
Jericho HTML Parser	Java
Html Agility Pack	C#
Validator.nu HTML Parser	Java
Nokogiri	Ruby
Gumbo	C
HtmlUnit	Java
html5ever	Rust
jsoup	Java
html.parser	Python
PHP Simple HTML DOM Parser	PHP
AVHTML	C++
HTML Monkey	C#
MyHTML	C
HTML Tidy	ANSI C
TagSoup	Java

BeautifulSoup

Python integrates numerous libraries and frameworks for Web scraping and data analytics. BeautifulSoup (URL: <https://www.crummy.com/software/BeautifulSoup>) is one such library that is widely used for data extraction and

logging of outcomes from multiple real-time channels.

The key benefits of BeautifulSoup include the following:

- Compatible with Python 2 as well as Python 3
- Fast and high-performance library

- Dynamic parsing and extraction of Web pages
- Deep searching, navigation and parsing of content on Web pages
- Real-time extraction of data without delay
- Dynamic storage of Web data using HTTP requests to the Web URL
- Data extraction and transformation into multiple formats including JSON, XML, CSV, TXT, XLS, XLSX, and many others

Table 3 presents the many instructions to install BeautifulSoup.

BeautifulSoup is one of the very powerful libraries of Python that are used to extract real-time data for market research, predictive analytics, and knowledge discovery, including user sentiments about products and their features, as well as reviews on new consumer products.

Scraping messages and user timelines from Twitter and other social media

We often need to analyse the user feedback and reviews from social media about particular products or services. Generally, people share their views and feedback about products on Twitter, Facebook, Instagram and other similar social platforms. Web scraping can be done to extract these messages from timelines so that the users' sentiments can be evaluated in order to improve services or product features.

Scraping images and timelines on Instagram

Instagram is a very popular platform for sharing multimedia content using a mobile based platform. The content that users broadcast on Instagram can be fetched using the BeautifulSoup library. One of the key applications of this implementation can be to evaluate the current status of a particular user so that his or her behaviour and interests can be identified.